Hardware Technical Reference

Nemo Winch V3.0

N20 Edition

Specialized Embedded Winch Controller for Autonomous Drone Payload Delivery

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Document Version:	1.0
Hardware Revision:	V3.0 - N20 Architecture
Board Type:	Embedded Payload Winch Controller PCB
Primary MCU:	Seeed Studio XIAO ESP32-C3
Motor Driver:	TB6612FNG Dual H-Bridge
Organization:	PSYC Aerospace and Defence Industries Pvt. Ltd.

Confidential Engineering Document

Prepared for internal design review, assembly, testing, and project documentation.

Contents

1 System Overview

The **Nemo Winch V3.0 N20 Edition** is a specialized embedded control PCB designed for autonomous drone payload delivery and controlled payload retrieval. This version is optimized around a compact **N20 DC gear motor** and a hybrid control architecture that separates high-speed real-time tasks from slower state-management functions.

The board interfaces with standard drone flight controllers such as Pixhawk-class autopilots, while maintaining local safety fallbacks, manual override capability, precision spool depth tracking, and isolated power delivery for logic and actuator domains.

1.1 Primary Functions

- Controlled deployment and rewind of a payload using an N20 DC gear motor.
- Flight-controller PWM command input for autonomous winch operation.
- Manual override for field testing, recovery, and safety intervention.
- Magnetic encoder-based spool position and payload depth tracking.
- Servo-driven clutch or payload release mechanism.
- UART telemetry output for depth, motor state, fault state, and mode status.
- Local docking/contact switch input for top-limit detection.

1.2 System Architecture Summary

Table 1: High-level system architecture

Subsystem	Implementation
Primary MCU	Seeed Studio XIAO ESP32-C3
Logic Expansion	PCF8574T I2C GPIO expander
Motor Driver	TB6612FNG dual H-bridge, one channel used for N20 motor
Depth Sensor	AS5600 magnetic rotary encoder
Power Input	XT60 battery input, up to 6S LiPo class input
5V Regulation	TPS54302 synchronous buck converter
Flight Controller Interface	Pixhawk PWM input and UART telemetry connector
Manual Control	SPDT manual switch header
Payload Release	5V servo output header
Safety Input	Contact terminal / docking switch input

2 Core Processing and Logic Architecture

The board uses a hybrid logic architecture. Time-critical signals are handled directly by the ESP32-C3, while non-time-critical control and safety-state inputs are delegated to an I2C GPIO expander.

2.1 Primary Microcontroller

Table 2: Primary MCU configuration

Parameter	Description
MCU Module	Seeed Studio XIAO ESP32-C3
Logic Level	3.3V
Primary Tasks	Motor PWM generation, Pixhawk PWM input capture, UART telemetry formatting, servo PWM generation, safety-state processing
Wireless Capability	Wi-Fi and BLE available through ESP32-C3, optional for future telemetry or diagnostics
Programming Interface	USB-C on XIAO module

2.2 ESP32-C3 Signal Responsibilities

Table 3: ESP32-C3 signal allocation

Signal	Function	Description
D0 / PWM Output	Motor Speed Control	Generates high-frequency PWM for TB6612FNG PWMA input.
Pixhawk PWM Input	Command Input	Reads RC/Pixhawk PWM command for release, hold, rewind, or stop logic.
Servo PWM	Clutch Servo	Drives the 5V servo signal line for the mechanical payload release or clutch system.
UART TX/RX	Telemetry	Sends depth and state data to Pixhawk or external telemetry bridge.
I2C SDA/SCL	Sensor and Expander Bus	Communicates with AS5600 encoder and PCF8574T GPIO expander.
Boot/Reset Lines	Programming and Debug	Used for ESP32-C3 flashing and recovery.

2.3 I/O Expansion Domain

The **PCF8574T** is used as the slow-logic I/O expander. It reduces direct GPIO usage on the ESP32-C3 and keeps state-based operations away from high-speed timing paths.

Table 4: PCF8574T slow-logic allocation

Expander Pin	Connected Function	Description
P0	TB6612 STBY	Enables or disables the motor driver standby state.
P1	TB6612 AIN1	Motor direction control input 1.
P2	TB6612 AIN2	Motor direction control input 2.
P3	Contact Switch	Reads docking or top-limit contact switch.
P4	Manual Switch 1	Reads manual UP / rewind side of SPDT switch.
P5	Manual Switch 2	Reads manual DOWN / release side of SPDT switch.
P6	Status LED / Spare	Optional visual indication or spare output.
P7	Spare / Future Use	Reserved for future expansion.

> Engineering Note
>
> The I2C bus should operate at standard mode, 100kHz. The SDA and SCL lines are pulled up to 3.3V using dedicated pull-up resistors. A value of 10kΩ is acceptable for short traces and low bus capacitance.

3 Power Distribution Network

The Nemo Winch V3.0 uses separated power domains to isolate sensitive logic circuitry from actuator noise generated by the motor and servo.

3.1 Power Input

Table 5: Main power input

Parameter	Description
Connector	XT60-M connector
Input Net	VBAT_IN
Battery Class	Up to 6S LiPo class input
Nominal Voltage	Approximately 22.2V to 24V for 6S LiPo operation
Protection	Fused input path for catastrophic short-circuit protection
Primary Conversion	TPS54302 buck regulator generates the 5V rail

3.2 5V Regulation

The board uses a **TPS54302 synchronous buck converter** to step the raw battery voltage down to a regulated 5V rail.

Table 6: 5V buck rail

Parameter	Description
Regulator IC	TPS54302
Output Voltage	5V
Output Current Capability	Up to 3A, layout and thermal conditions dependent
Loads Powered	N20 motor VM rail, clutch servo, ESP32 module input regulator, optional external 5V accessories
Rail Name	+5V

3.3 3.3V Logic Rail

The ESP32-C3 module provides or distributes the 3.3V logic domain. All logic devices connected to the ESP32-C3 must operate at 3.3V-compatible logic levels.

Table 7: 3.3V logic rail loads

Load	Description
ESP32-C3 Logic	Internal microcontroller domain
PCF8574T	I2C GPIO expander
AS5600	Magnetic angle sensor, when operated in 3.3V mode
TB6612FNG VCC	Logic supply for motor driver input compatibility
I2C Pull-ups	SDA and SCL pulled up to 3.3V

3.4 Transient Voltage Suppression

A dedicated **SMAJ5.0A TVS diode** is placed across the 5V and GND rail near the motor driver.

Engineering Note

The TVS diode acts as a fast transient clamp. It helps absorb inductive spikes and rail disturbances caused by the brushed N20 motor during hard braking, rapid direction reversal, or sudden motor disconnection.

Safety Warning

The TVS diode is not a replacement for correct motor decoupling, short return paths, and proper ground routing. The motor driver VM capacitor should be physically close to the TB6612FNG VM and GND pins.

4 Motor Control Subsystem

4.1 Motor Driver

The board uses a **TB6612FNG dual H-bridge motor driver**. One H-bridge channel is used to control the N20 DC gear motor.

Table 8: TB6612FNG motor driver configuration

Parameter	Description
Driver IC	Toshiba TB6612FNG
Motor Channel Used	Channel A
Motor Output Nets	AO1 and AO2
Motor Supply	VM connected to regulated 5V rail
Logic Supply	VCC connected to 3.3V logic rail
Speed Control	PWMA driven directly by ESP32-C3 PWM
Direction Control	AIN1 and AIN2 driven by PCF8574T expander
Standby Control	STBY driven by PCF8574T expander

4.2 Motor Driver Truth Table

Table 9: TB6612FNG channel-A control logic

STBY	AIN1	AIN2	PWM	Motor State
0	X	X	X	Standby / driver disabled
1	1	0	PWM	Forward / release direction
1	0	1	PWM	Reverse / rewind direction
1	0	0	X	Short brake or stop, depending firmware policy
1	1	1	X	Short brake

4.3 Motor Output Connector

Table 10: N20 motor output connector

Pin	Net	Description
1	AO1	Motor terminal 1
2	AO2	Motor terminal 2

4.4 Motor Decoupling

The motor driver VM pin should be locally decoupled with a bulk capacitor and high-frequency ceramic capacitor.

Table 11: Recommended motor-driver decoupling

Component	Typical Value	Placement Requirement
Bulk Capacitor	10μF or higher	Close to TB6612FNG VM and GND
Ceramic Capacitor	0.1μF	As close as possible to VM pin
TVS Diode	SMAJ5.0A	Across 5V and GND near motor driver

5 Sensors and Telemetry

5.1 AS5600 Magnetic Rotary Encoder

The payload depth is measured using an **AS5600 magnetic rotary encoder**. The encoder observes a diametrically magnetized magnet mounted coaxially on the N20 spool shaft.

Table 12: AS5600 encoder configuration

Parameter	Description
Sensor	AS5600 magnetic rotary encoder
Interface	I2C
Default I2C Address	0x36
Resolution	12-bit absolute angle measurement
Measured Quantity	Spool shaft angle
Derived Quantity	Spool rotations and payload deployment depth
Supply	3.3V logic rail recommended

5.2 Depth Calculation

The ESP32-C3 calculates payload deployment depth by tracking angular displacement and converting spool rotation into linear cable payout.

$$\text{Depth} = N_{\text{rotations}} \times \pi \times D_{\text{effective spool}}$$

Where:

- $N_{\text{rotations}}$ is the total counted spool rotation count.
- $D_{\text{effective spool}}$ is the effective winding diameter of the spool.
- The effective diameter should be calibrated because cable layering changes the real payout diameter.

Engineering Note

For accurate field operation, perform a calibration routine by deploying a known cable length and adjusting the effective spool diameter constant in firmware.

5.3 UART Telemetry

The board provides UART telemetry to the flight controller or external telemetry system.

Table 13: Pixhawk telemetry connector

Pin	Signal	Description
1	TX	ESP32-C3 transmit to Pixhawk RX
2	RX	ESP32-C3 receive from Pixhawk TX
3	GND	Common ground reference

5.4 Suggested Telemetry Fields

Table 14: Recommended telemetry packet fields

Field	Description
Mode	AUTO, MANUAL_UP, MANUAL_DOWN, HOLD, FAULT
Depth	Estimated deployed cable length
Motor State	FORWARD, REVERSE, BRAKE, OFF
PWM Command	Current motor PWM duty cycle
Limit State	Contact switch active or inactive
Encoder Status	AS5600 connected, magnet detected, angle valid
Fault Code	Over-limit, encoder error, command timeout, or power fault

6 Safety Interlocks and External Interfaces

6.1 Contact Terminal

The contact terminal is used as a docking or top-limit switch. It can be connected to a mechanical limit switch, conductive ring, or contact pad on the winch housing.

Table 15: Contact terminal connector

Pin	Signal	Description
1	Contact Input	Routed to PCF8574T input pin
2	GND	Ground reference for switch closure

Safety Warning

The contact terminal should be treated as a safety input. When the payload hook reaches the top docking point, the firmware should immediately cut motor PWM and command a safe motor state.

6.2 Manual Override Header

The manual override input allows field operators to force rewind or release independent of automatic Pixhawk control.

Table 16: Manual override switch positions

Switch Position	Electrical State	Function
UP	Manual UP pin bridged to GND	Manual rewind override
CENTER	Both inputs floating HIGH	Auto flight-controller mode
DOWN	Manual DOWN pin bridged to GND	Manual release or clutch override

6.3 Pixhawk Control Input Header

The Pixhawk control input header receives the primary mission command from the flight controller.

Table 17: Pixhawk control input header

Pin	Signal	Description
1	GND	Ground reference
2	PWM1	Pixhawk AUX PWM input to ESP32-C3
3	PWM2 / Spare	Optional second command input or spare
4	ZERO RESET	Optional encoder/depth zero reset command

6.4 Clutch Servo Output

The board provides a standard 3-pin servo header for a clutch or payload release actuator.

Table 18: Servo output header

Pin	Signal	Description
1	+5V	Servo power from 5V rail
2	PWM	Servo command from ESP32-C3
3	GND	Servo ground

Safety Warning

Servo stall current can cause 5V rail dips. If the clutch servo is high torque, validate the 5V rail under worst-case servo stall and motor startup together.

7 Connectors and Ports

Table 19: Board connector summary

Connector	Interface	Function
XT60-M	Power Input	Main VBAT input from drone battery.
N20 Motor Out	2-pin Motor Output	Connects TB6612FNG AO1/AO2 to N20 motor terminals.
Pixhawk Input Header	PWM Input	Receives flight-controller command PWM.
Pixhawk Telemetry	UART	Sends winch depth and status telemetry.

Connector	Interface	Function
AS5600 Connector	I2C Sensor	Connects magnetic rotary encoder module.
Contact Terminal	Switch Input	Top-limit or docking switch.
Manual Switch Header	SPDT Input	Manual rewind, auto, or release command.
Servo Header	3-pin Servo	Drives clutch or payload release servo.
Power Selection Port	Power Routing	Allows external 5V or buck-derived 5V selection depending assembly option.
External DC In-/Out	Auxiliary Power	Allows external power routing depending system integration.

8 Firmware Control Logic

8.1 Operating Modes

Table 20: Recommended operating modes

Mode	Description
BOOT	Initialize GPIO, I2C, encoder, PWM timers, and UART telemetry.
SAFE HOLD	Motor disabled or braked; waits for valid command.
AUTO RELEASE	Pixhawk command deploys payload using controlled PWM.
AUTO REWIND	Pixhawk command rewinds payload until contact switch or target depth is reached.
MANUAL UP	Manual switch forces rewind.
MANUAL DOWN	Manual switch forces release or clutch action.
FAULT	Motor PWM disabled due to sensor fault, timeout, over-limit, or invalid state.

8.2 Recommended Safety Priorities

The firmware should evaluate safety conditions in the following priority order:

1. Emergency fault state.
2. Contact terminal active during rewind.
3. Manual override command.
4. Pixhawk command validity and timeout.
5. Encoder validity and magnet detection.
6. Normal automatic motor command.

8.3 Motor Command Policy

Table 21: Suggested motor command policy

Command	Motor State	Notes
Stop	PWM = 0, brake or standby	Use brake for short hold, standby for low power.
Release	Forward PWM	Direction depends on physical motor polarity.
Rewind	Reverse PWM	Stop immediately if contact terminal is active.
Fault	PWM = 0, STBY = 0	Disable motor driver until fault cleared.
Manual Override	Direct operator mode	Still respect contact limit and critical faults.

9 Electrical Verification Checklist

9.1 Pre-Power Checks

Table 22: Pre-power inspection checklist

Check	Expected Result
VBAT to GND resistance	No short circuit. Resistance should not be near 0Ω.
5V to GND resistance	No short circuit.
3.3V to GND resistance	No short circuit.
TB6612FNG orientation	Pin 1 and package orientation verified.
TPS54302 orientation	Regulator IC and diode/inductor path verified.
TVS diode polarity	Correctly placed across 5V and GND.
I2C pull-ups	SDA and SCL pulled to 3.3V, not 5V.
Motor output	AO1/AO2 not shorted to GND or 5V.
Servo header polarity	+5V, signal, and GND order verified.

9.2 Power-Up Checks

Table 23: Power-up validation checklist

Check	Expected Result
5V buck output	5.0V nominal before connecting motor and servo.
3.3V logic rail	3.3V nominal.
ESP32-C3 boot	USB serial visible and firmware boots normally.
I2C scan	PCF8574T detected at 0x20 and AS5600 detected at 0x36.
Motor idle state	Motor remains off during boot.
Manual switch readout	UP, CENTER, DOWN states correctly detected.
Contact terminal	Switch closure detected reliably.
PWM input	Pixhawk PWM pulse width measured correctly.
Servo output	Servo moves only when commanded.

9.3 Motor Test Procedure

1. Power the board from a current-limited bench supply.
2. Start at a low current limit and verify 5V and 3.3V rails.

3. Upload motor test firmware with PWM duty limited to 20%.
4. Confirm TB6612FNG STBY, AIN1, AIN2, and PWMA logic levels.
5. Connect the N20 motor.
6. Command forward direction at low PWM.
7. Command stop.
8. Command reverse direction at low PWM.
9. Trigger the contact terminal and verify immediate motor cutoff.
10. Test manual UP and DOWN commands.

Safety Warning

Do not test motor direction reversal at high PWM during first bring-up. Brushed DC motors can generate large current transients during rapid reversal.

10 Mechanical Integration Notes

10.1 AS5600 Magnet Alignment

For reliable encoder operation, the magnet must be mounted coaxially with the AS5600 sensor.

- Use a diametrically magnetized magnet.
- Keep the magnet centered above the sensor.
- Maintain the air gap recommended by the AS5600 module or IC datasheet.
- Avoid ferromagnetic screws or brackets near the sensor if they distort the magnetic field.

10.2 Cable Spool Calibration

Cable payout depth is affected by changing spool diameter as cable layers build up or unwind. For accurate depth tracking:

1. Mark a known cable length, for example 1m.
2. Deploy the cable slowly.
3. Record encoder rotation count.
4. Calculate effective spool circumference.
5. Store calibration constant in firmware.

11 Recommended Firmware Constants

Listing 1: Suggested firmware configuration constants

```
#define I2C_ADDR_PCF8574        0x20
#define I2C_ADDR_AS5600         0x36

#define MOTOR_PWM_FREQ_HZ       20000
#define MOTOR_PWM_MIN_DUTY      0
#define MOTOR_PWM_MAX_DUTY      255
```

```
#define PIXHAWK_PWM_MIN_US      1000
#define PIXHAWK_PWM_MID_US      1500
#define PIXHAWK_PWM_MAX_US      2000
#define PIXHAWK_TIMEOUT_MS      500

#define SERVO_PWM_FREQ_HZ       50
#define SERVO_RELEASE_US        1900
#define SERVO_LOCK_US           1100

#define CONTACT_ACTIVE_LOW      1
#define MANUAL_ACTIVE_LOW       1
```

12 Known Design Considerations

Table 24: Important design considerations

Area	Recommendation
Motor Noise	Keep TB6612FNG, VM capacitor, and motor connector loop area as small as possible.
Grounding	Use a solid ground plane and avoid routing encoder/I2C traces near motor current loops.
Servo Current	Validate 5V rail for servo stall current and motor startup current together.
I2C Robustness	Keep I2C traces short and pulled up only to 3.3V.
Thermal	Check TPS54302 and TB6612FNG temperature under continuous motor operation.
Firmware Safety	Default motor state must be OFF during boot, reset, and communication loss.
Connector Polarity	Clearly mark servo, motor, battery, and Pixhawk connector orientation on silkscreen.

13 Revision History

Table 25: Document revision history

Version	Date	Notes
1.0	Initial Release	First complete technical reference for Nemo Winch V3.0 N20 Edition.

Disclaimer

This document is intended for engineering design review, assembly guidance, and testing support. Final electrical values, PCB layout decisions, protection ratings, and flight-readiness validation must be verified through bench testing, environmental testing, and system-level integration testing before field deployment.

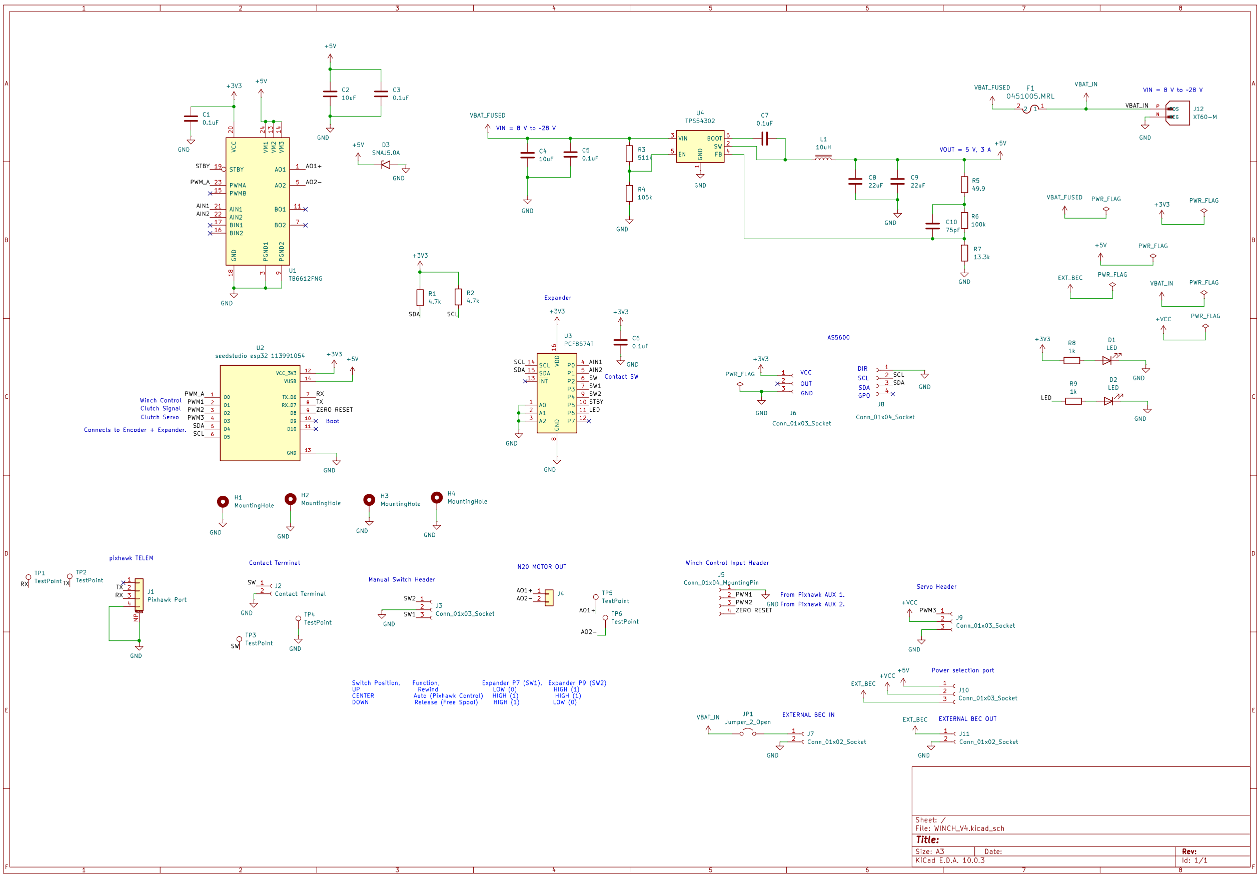

Figure 1: Schematic